

Execute Decision Flow with SQL Query Node in SAS Studio

Decision flows in Intelligent Decisioning that access a database cannot be published to CAS because the ds2 sqlstmt package which is used to access the database is currently not supported in CAS. However, but it is supported in SPRE. Therefore, we can export the ds2 decision code from Intelligent Decisioning and execute it in SPRE from SAS Studio.

The ds2 macro described here will extract the decision code from Intelligent Decisioning, prepare it to take an input table and to write the result to an output table analogue to the Intelligent Decisioning macro %DCM_EXECUTE_DECISION.

The macro is developed and tested with Intelligent Decisioning 5.3.

Install %DCM_EXECUTE_DECISION

- Move file *get_decision_code_SPRE.py* to Viya server.
 - Create directory: /etc/sas/dec-spre/app
 - `sudo mkdir /etc/sas/dec-spre/app`
 - `sudo chmod 755 /etc/sas/dec-spre/app`
 - Create directory: /etc/sas/dec-spre/dec-ds2code
 - `sudo mkdir /etc/sas/dec-spre/dec-ds2code`
 - `sudo chmod 777 /etc/sas/dec-spre/dec-ds2code`
 - Move file *get_decision_code_SPRE.py* to directory /etc/sas/dec-spre/app
- Set Viya user credentials in file *get_decision_code_SPRE.py*
 - Open file
 - Set values for variables: uid, pwd, clientId, clientSecret
 - uid: User to logon to Viya
 - pwd: User password
 - clientId: Client Id needed to get access_token to call Viya web services
 - clientSecret: Client Secret needed to get access_token to call Viya web services

Note: The macro requires the DS2 PyMAS Package to be configured for Workspace server and SAS Cloud Analytic Services. See [documentation](#) for more information.

For information to register ClientId and ClientSecret see Registering Clients [documentation](#).

- Save macro to macro library
 - Upload file *CallDecisionInSPRE_macro.sas* into SAS Studio
 - Open file *CallDecisionInSPRE_macro.sas*
 - Run code to create and save macro

Syntax

```
%DCM_EXECUTE_DECISION_SPRE (URI=decision-URI,  
    INPUTTABLE=<libref.>table,  
    OUTPUTTABLE=<libref.>table
```

<, optional-arguments>
)

Required Arguments

URI=decision-URI

specifies the URI of the decision that you want to execute. This can be a decision or decision-revision uri.

INPUTTABLE=<libref.>table

specifies the input table for the decision to process. If you specify a single-level name, the macro looks for the table in the WORK library.

OUTPUTTABLE=<libref.>table

specifies the name of the output table that is created by the decision. If you specify a single-level name, the macro writes the table to the WORK library.

Optional Arguments

LIBS=<libref ... >

Specifies a list of libraries (blank delimited) accessing database schemas. There needs to be a library name for each database schema accessed in a SQL Query node.

Default: WORK

Note: The macro is executed by the workspace server and will process data from and to SAS 9 libraries. If data to process is in a CAS library, it needs to be loaded into a SAS 9 library first.

Example

```
/*Assign the macro library where the macro is stored*/
%if (%sysfunc(libref(MACROS))) %then %do;
    libname macros '/opt/sas/spre/config/etc/macros';
    options mstored sasmstore=macros;
%end;

%let URI= /decisions/flows/a2de1494-038b-4ab3-9c81-0d2f5ca4e5b4/revisions/c47eb35f-9d49-4152-af58-b84d07a5709c;

/*Set database connection for the database the decision flow will access*/
libname PGDATA postgres server=localhost port=5433 user=sasdemo password='123456' database=postgres schema=PGDATA;

/*Execute decision flow through macro*/
%DCM_EXECUTE_DECISION_SPRE (URI= &URI,
    INPUTTABLE= work.test_truck,
    OUTPUTTABLE= work.test_truck_out,
    LIBS= PGDATA);
```